A Graph Convolutional Network for Detecting Parkinson's Disease Using Brain Functional Connectivity from fMRI data

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ECE 285
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Team Introduction

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Project Summary

1

Problem

- Understanding Functional Connections in the Pathological Brain

2

Methods

- Data Preprocessing
- Extract BOLD Signals
- Connectivity Matrix
- Train GCN Model
- Optimize

3

Results

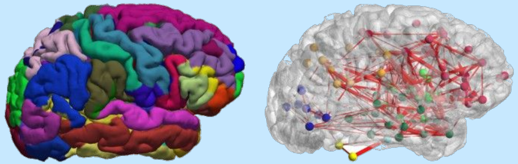
- We know there is something meaningful about the connectivity in PD.

<u>Key Words</u>: Functional Magnetic Resonance Image (**fMRI**), Blood-Oxygenation-Level-Dependent (**BOLD**) Signals, Graph Convolutional Network (**GCN**), Parkinson's Disease, MRI Segmentation

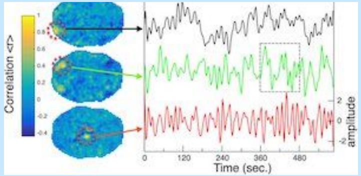
Introduction

Background and Motivation

1. Functional Connectivity: The activity patterns of anatomically distinct brain regions that are connected over time



2. Blood Oxygen Level Dependent Signal (BOLD) Signal: Links neuronal activity during information processing & MRI signal strength.



3. Many pathologies are a result of brain circuitry dysfunction which are not illuminated by structural MRI.

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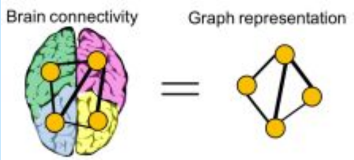
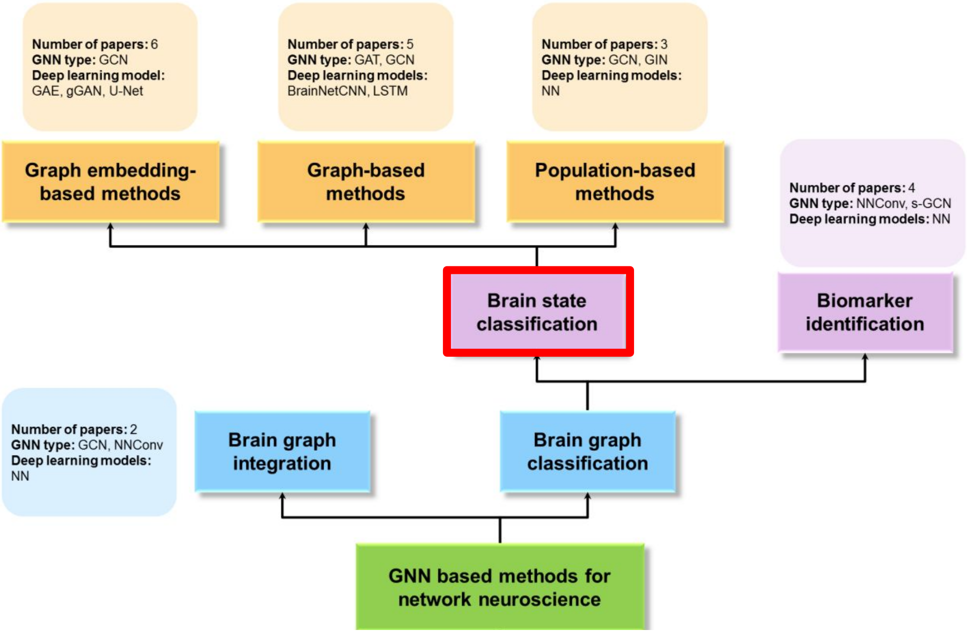
We use fMRI to probe networks and compute a “**functional connectome**”.

4. Neuroscientists have found that the brain can be modeled as a graph. 💡 **Can we learn a characteristic functional connectivity pattern in Parkinson’s Disease?**



Related Works

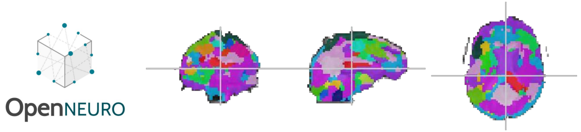
- Few (14) papers on **brain state classification** with GCN in last few years (especially Parkinson's)
- Recent techniques using fMRI aim to understand brain networks in greater detail, but...
- Current machine learning methods are more focused on MRI features and ignore brain connectivity



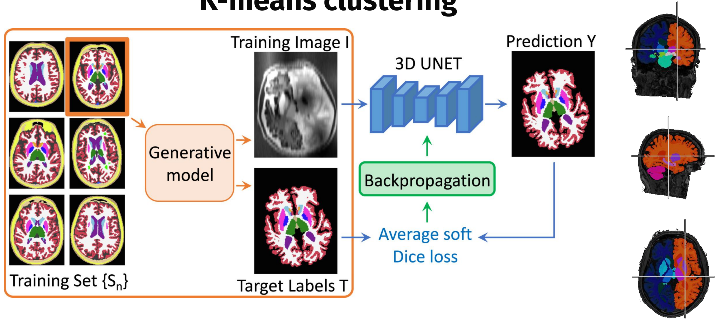
Methodology

Workflow

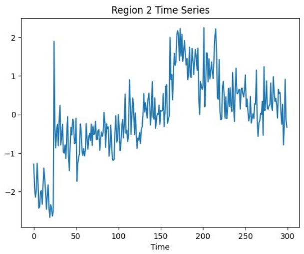
Data Processing

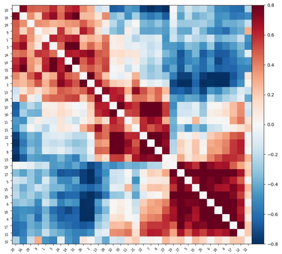


K-means clustering



SynthSeg

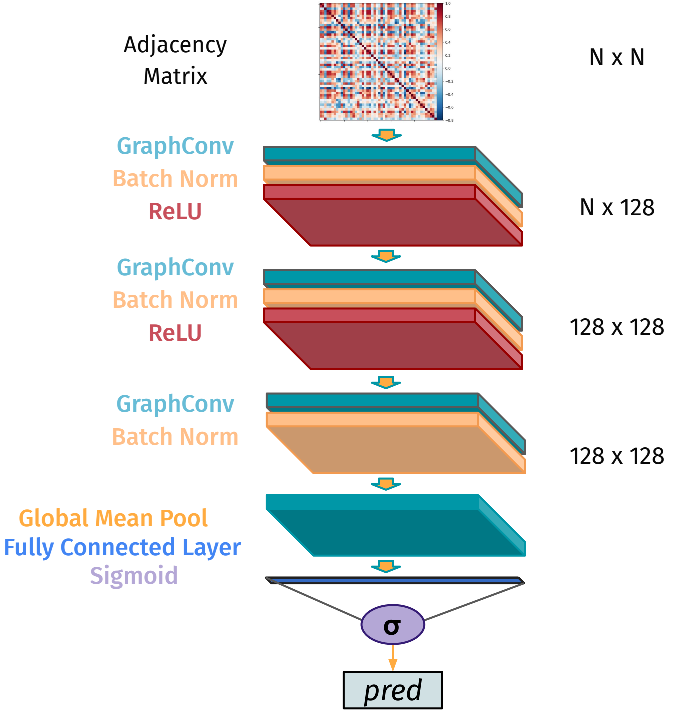




Time Series

Adjacency Matrix (AM)

GCN Model



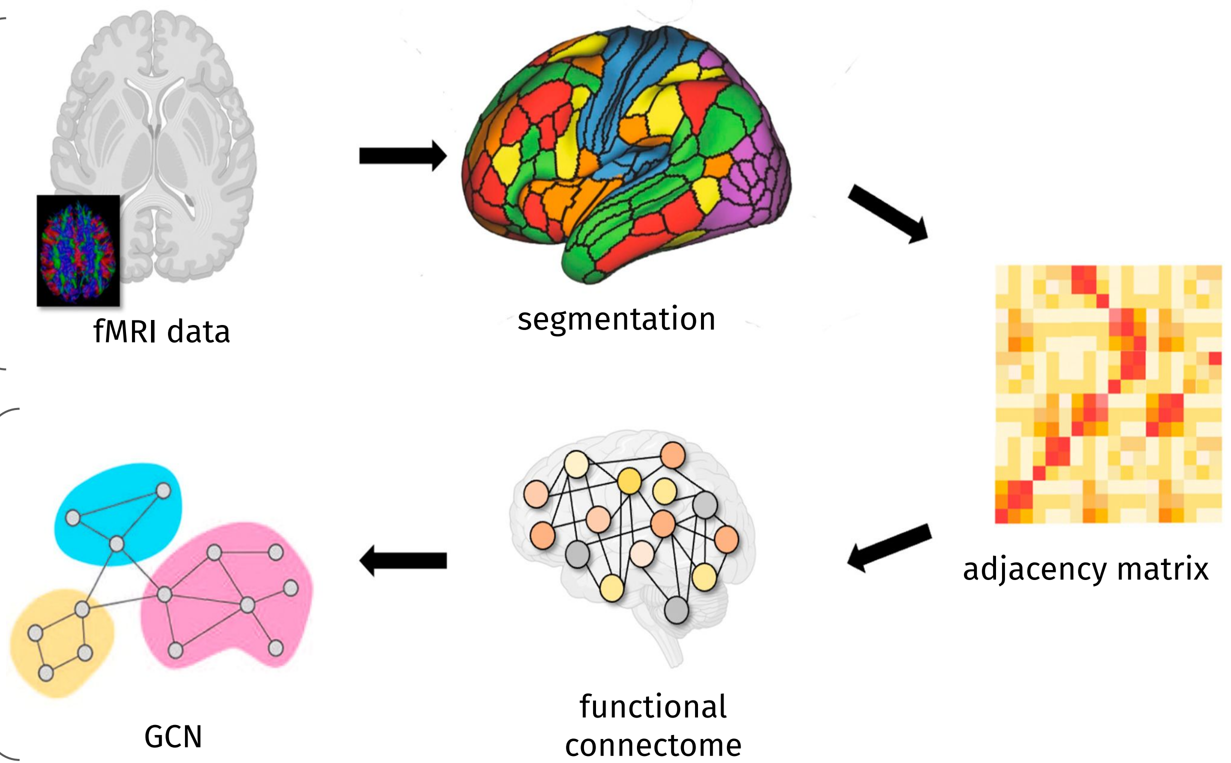
$$BCE = -\frac{1}{N}\sum_{i=0}^{N} y_i \cdot log(\hat{y}_i) + (1 - y_i) \cdot log(1 - \hat{y}_i)$$

$$h_i^{(l+1)} = \sigma(b^{(l)} + \sum_{j \in \mathcal{N}(i)} \frac{1}{c_{ji}} h_j^{(l)} W^{(l)})$$

Highlights of Methods

Spatio-temporal physical application
- Temporal: dynamics of brain activity
- Spatial: functional interactions between brain regions

Graph Convolutional Network
- Addresses both spatial and temporal contributions to functional connectivity
- Trained with convolutional operator on graphical representation of BOLD time series
- Derives edge importance between features
- Identifies selective functional connections in brain significant to disease and normal states
- Leverages information processing technique



Experiments

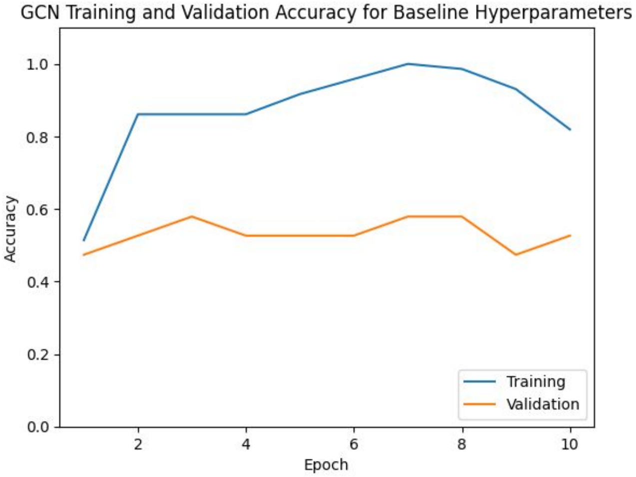
Experiment 1: Architecture and Hyperparameters (Baseline)

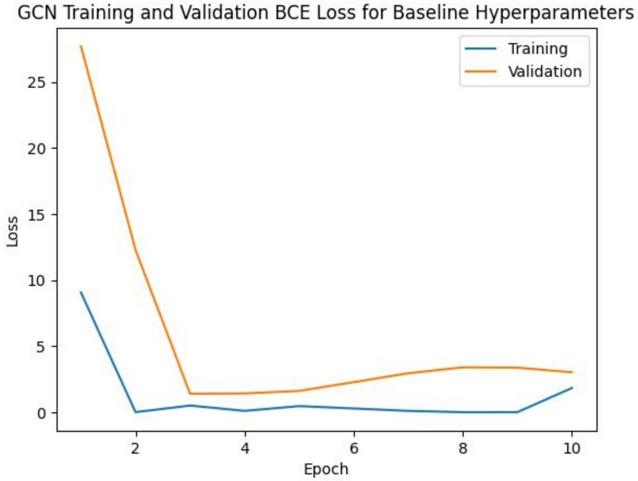
Batch Normalization: Yes

# of GCN Layers: 3

Regularization: None

Fixed Hyperparameters





Train Acc: 100%

Validation Acc: 58%

Training Loss: 0.0056

Validation Loss: 1.4

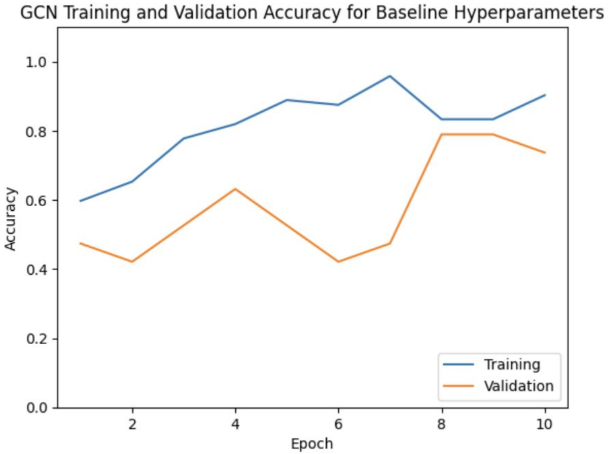
Experiment 1: Architecture (Optimized)

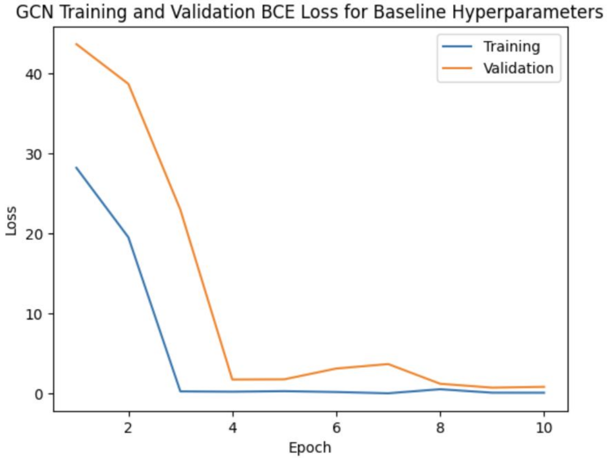
Batch Normalization: Yes

# of GCN Layers: 4

Regularization: Dropout

Fixed Hyperparameters





Train Acc: 90.28 %

Validation Acc: 73.68%

Training Loss: 0.0696

Validation Loss: 0.8086

Experiment 1: Hyperparameters (Optimized)

Graph Complexity: 20

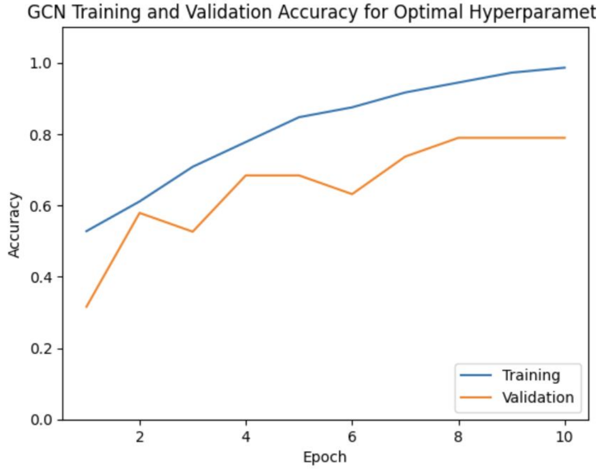
Batch Size: 22

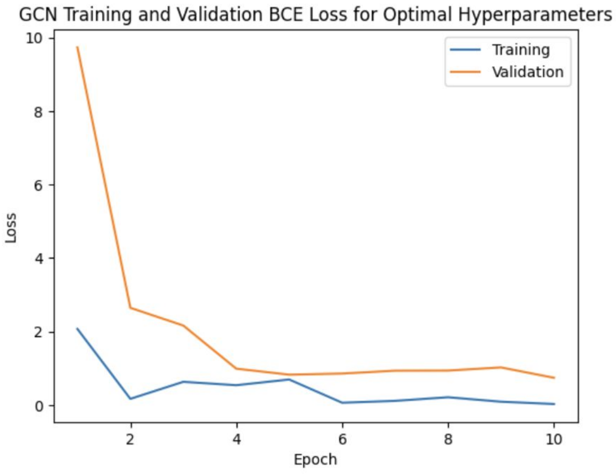
Hidden Channels: 128

Threshold: 0.6

Learning Rate: 1e-3

Epochs: 10





Train Acc: 98.61%

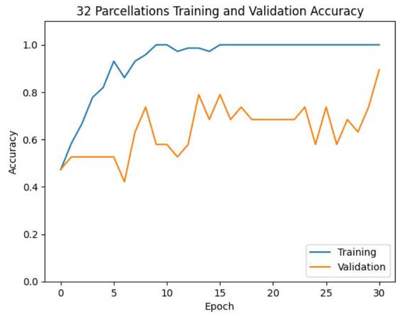
Validation Acc: 78.95%

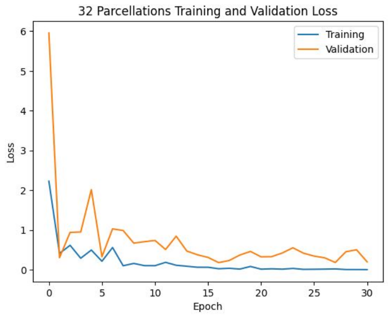
Training Loss: 0.0310

Validation Loss: 0.7442

Experiment 2: Segmentation Methods

K-means clustering: 32 Parcellations





TA: 100%

TL: .008

VA: 89.47%

VL: .178

Graph Complexity: 20

Hidden Channels: 128

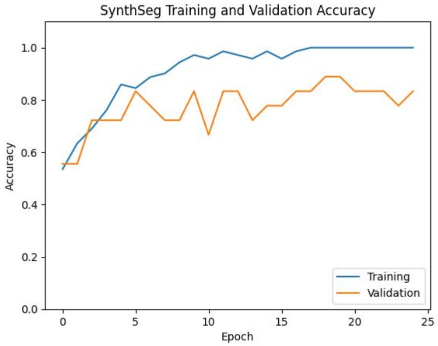
Batch Size: 16

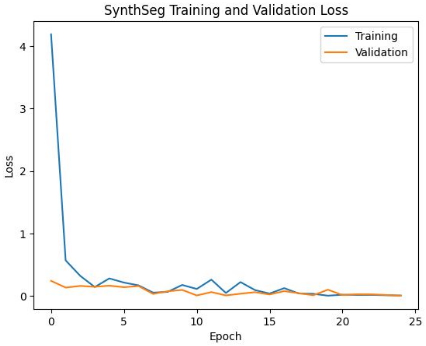
Threshold: 0.6

Learning Rate: 1e-3

Epochs: 30

SynthSeg: 32 Regions





TA: 100%

TL: 0.002

VA: 88.9%

VL: .0044

Graph Complexity: 20

Hidden Channels: 128

Batch Size: 16

Threshold: 0.6

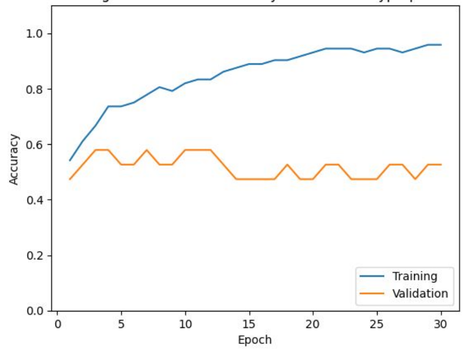
Learning Rate: 1e-3

Epochs: 24

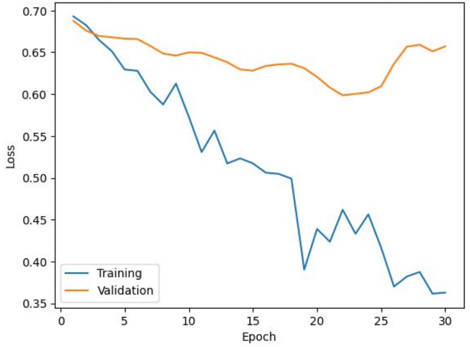
Experiment 3: Model in Literature

Baseline (TA: 95%, VA: 57%, TL: 0.36, VL: 0.65)

GCN Training and Validation Accuracy for Baseline Hyperparameters

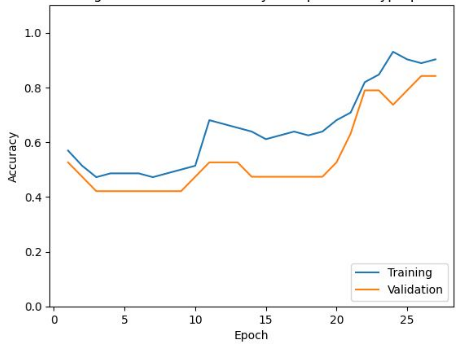


GCN Training and Validation BCE Loss for Baseline Hyperparameters

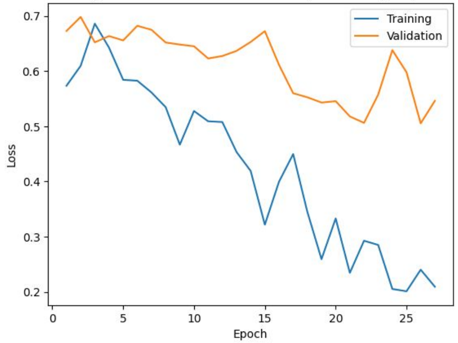


Optimized (TA: 95%, **VA: 84%**, TL: 0.24, VL: 0.52)

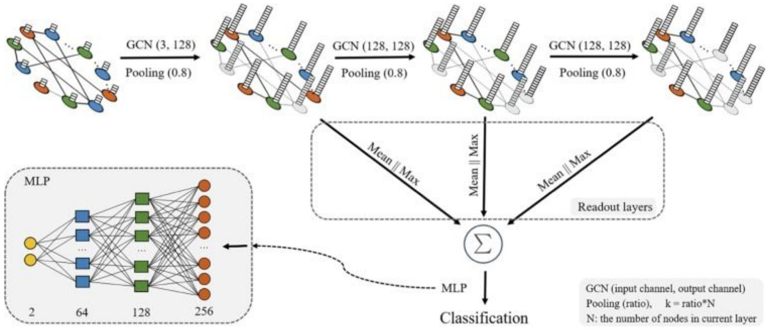
GCN Training and Validation Accuracy for Optimized Hyperparameters



GCN Training and Validation BCE Loss for Optimized Hyperparameters



Classification of schizophrenia patients using a graph convolutional network: A combined functional MRI and connectomics analysis



Increased Dropout Probability (0.8 vs 0.5)

Larger Learning Rate (1e-3 vs 1e-4)

Larger Batch Size (64 vs 10)

Discussion

Conclusions, Limitations, and Future Works

Conclusions

- Parkinson's is more accurately detected with **less parcellations and nodes**
- **Meaningful pattern of connection** between different brain regions attributed to Parkinson's
- Functional connectivity of different neurological disorders may exhibit **varying complexity**

Limitations

- Small dataset → training and validation inconsistencies
- Generalizability
- Dataset not used in literature

Future Works

- Implement larger dataset and more complex model
- Expand towards different neurological disorders
- Experiment with other models in literature

What Have We Learned?

✘ Deep learning segmentation methods for this GCN model

✔ Graphical representation of functional connectivity from BOLD time series signals for insight into diseased vs. healthy state differences

✔ Dropout regularization for smaller dataset & complex networks

✔ GCN residual models for classification

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